

Jean-Emmanuel Kouadio

emmanuelkj5@gmail.com | 681-283-0298 | [linkedin/j-emanuel-k](https://www.linkedin.com/in/j-emanuel-k/) | <https://jeanemmanuelk.github.io/portfolio/>

EDUCATION

M.S. Data Analytics and Information Systems

Jan. 2022 - Dec. 2024

Shepherd University, Shepherdstown, WV

GPA: 3.73/4.0

Relevant Coursework: Machine Learning, Big Data Analytics, Statistical Analysis, Business Analytics, Database Systems

B.S. Computer Engineering

Aug. 2017 - Dec. 2021

Shepherd University, Shepherdstown, WV

GPA: 3.45/4.0

Research Interests: Machine Learning and Artificial Intelligence, Healthcare and Biomedical Data Science, Applied Deep Learning for Medical Decision Support, Interpretable and Trustworthy Artificial Intelligence

RESEARCH EXPERIENCE

Research Assistant - West Virginia INBRE (NIH-funded Program)

Dec. 2018 - Jul. 2021

Shepherd University | Shepherdstown, WV

- Conducted biomedical cancer research investigating mechanisms driving metastatic lung cancer tumor growth, supporting the development of improved therapeutic strategies.
- Performed quantitative and statistical analysis of experimental and clinical datasets to evaluate immunotherapy and radiation therapy combinations, identifying effective dosage thresholds and improving treatment effectiveness by approximately 15%.
- Developed and analyzed mathematical and computational models using systems of impulsive ordinary differential equations (ODEs) to simulate immune–tumor interactions and treatment response.
- Analyzed treatment response and survival outcome data to support data-driven decision-making in cancer research.
- Presented research findings at regional scientific conferences (PWVAS), communicating results to audiences of over 200 technical and non-technical participants.

Research Advisor: Dr. Qing Wang

PUBLICATIONS AND CONFERENCE PRESENTATIONS

Kouadio, J.-E., Wang, Q. (2019).

Modeling the effect of a combinatorial anti-CTLA antibody and radiation therapy on metastatic melanoma.

Proceedings of the West Virginia Academy of Science, 91(1), 94th Annual Meeting Abstracts.

Poster presentation.

<https://doi.org/10.55632/pwvas.v91i1.581>

(Supported by NIH Grant P20GM103434 – West Virginia IDeA Network for Biomedical Research Excellence)

Kouadio, J.-E., Wang, Q. (2020).

Modeling the effect of a combinatorial anti-CTLA antibody and radiation therapy on metastatic lung cancer.

Proceedings of the West Virginia Academy of Science, 92(1), 95th Annual Meeting Abstracts.

Poster presentation.

<https://doi.org/10.55632/pwvas.v92i1.708>

(Supported by NIH Grant P20GM103434 – West Virginia IDeA Network for Biomedical Research Excellence)

Kouadio, J.-E., Wang, Q. (2021).

A combinatorial anti-CTLA antibody and radiation therapy on metastatic lung cancer.

Proceedings of the West Virginia Academy of Science, 93(1), 95th Annual Meeting Abstracts (Virtual).

Poster presentation.

<https://doi.org/10.55632/pwvas.v93i1.827>

(Supported by NIH Grant P20GM103434 – West Virginia IDeA Network for Biomedical Research Excellence)

Guzide, O., Kouadio, J.-E. (2021).

The use of artificial intelligence in security system and everyday life and creating a security system using basic inexpensive devices (Raspberry Pi).

Proceedings of the West Virginia Academy of Science, 93(1), 95th Annual Meeting Abstracts (Virtual).

Oral presentation.

<https://doi.org/10.55632/pwvas.v93i1.785>

WORK EXPERIENCE

Data Analytics and Customer Service Intern

May 2026 - Present

SU Wellness Center | *Shepherdstown, WV*

- Provided front-line customer service support to members and guests, maintaining accurate records in the customer service database.
- Assisted in creating, distributing, and analyzing member satisfaction surveys, using data analytics tools to track engagement and program usage trends.
- Supported special projects focused on operational efficiency and member experience, collaborating with staff to ensure smooth daily operations.

Data Analyst

May 2025 - Jan. 2026

SU Wellness Center | *Shepherdstown, WV*

- Analyzed engagement and retention trends for 3,000+ students and community members across wellness programs, fitness facilities, and aquatic services.
- Developed real-time Power BI dashboards and defined membership KPIs, resulting in a 25% growth in renewal rates and improved data-driven decision-making.
- Built churn prediction models using Python, enabling targeted outreach that prevented over 100 member losses and supported data-driven intervention strategies through predictive modeling.

Data Scientist (Intern)

Jun. 2024 - Feb. 2025

Abt Global, Inc. | *Rockville, MD*

- Developed a machine learning-powered self-service tool to identify compensation disparities (85% accuracy), enabling HR and leadership to make data-driven decisions for 3,000+ employees.
- Automated the model lifecycle with Python and Oracle FDI; resolved 50+ data quality issues and improved HR analytics reliability by 20% through rigorous validation and testing.
- Built ETL pipelines and SQL-based dashboards that cut analysis turnaround time by 50%, enabling real-time insights and earning cross-team recognition.
- Presented insights through Power BI dashboards to 200+ stakeholders, translating complex data into high-level recommendations that informed cross-functional decision-making and accelerated policy changes.

Data Reporting Assistant (Member Services & Sales Analytics)

Jun. 2022 - Jul. 2024

Shepherd University | *Shepherdstown, WV*

- Built automated weekly and monthly reports tracking sales and membership trends using Excel and SQL.
- Supported data entry, cleaning, and validation in internal CRM systems, improving report accuracy by 30%.

TECHNICAL SKILLS

Programming Languages: Python, SQL, PySpark

Statistical & Machine Learning: Predictive Modeling, Classification, Regression, Time Series Analysis, A/B Testing, Hypothesis Testing, Feature Engineering, Model Validation, Exploratory Data Analysis

Big Data & Cloud: AWS (EC2, S3, SageMaker), Hadoop, Apache Spark, Oracle Cloud, Azure (Databricks, Data Factory)

Databases & ETL: SQL Server, Oracle, ETL Pipeline Development, Data Warehousing

ML Libraries & Frameworks: Scikit-learn, TensorFlow, Keras, XGBoost, Pandas, NumPy, SciPy, NLTK

Data Visualization & BI: Tableau, Power BI, Matplotlib, Seaborn, Oracle Analytics Cloud

Industry Knowledge: Credit Risk Analytics, Lending Analytics, Financial Modeling, Statistical Analysis

Certifications: Google Data Analytics Professional Certificate, Machine Learning Specialization (Stanford/Coursera)

Research Methods: Mathematical modeling (ODEs, impulsive systems), statistical analysis, simulation, model calibration, scientific communication